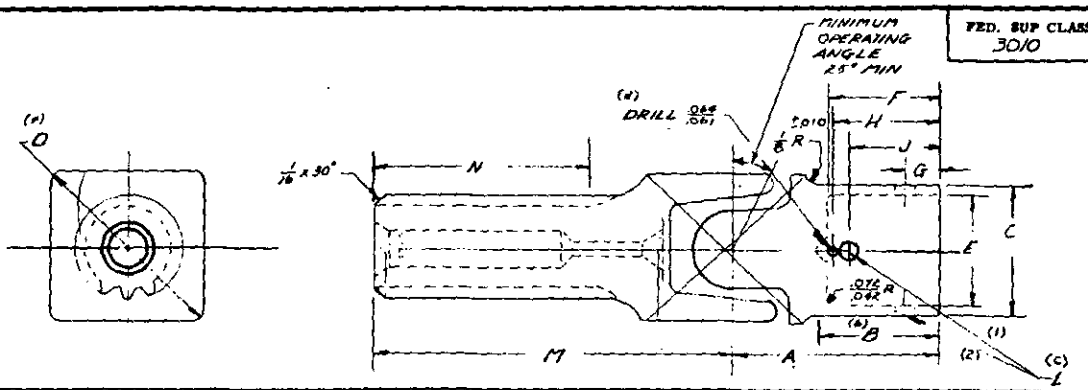
FED. SUP CLASS. 3010



DIMENSIONS-ROUND HUB

MS PART NO.	(a) FIIN	NOM SIZE	A	B	C DIA +.000 -.002	L SWING DIA	E DIA +.004 -.001	F	G	H	J	L PILOT HOLE DRILL
MS24314-12		3/4	1.313	1/16	.745	1 11/16	.625	1 3/16	.219	.930 .932	.531	5/32
MS24314-16		1	1.750	1 1/16	.995	1 11/16	.812	1	.281	.937 .939	.656	7/32
MS24314-20		1 1/4	2.188	1 1/16	1.245	1 3/4	1.062	1 1/16	.312	1.005 1.008	.688	7/32
MS24314-24		1 1/2	2.312	1 1/16	1.495	2 1/16	1.250	1 3/16	.344	1.180 1.183	.781	9/32
MS24314-28		1 3/4	2.563	1 1/16	1.745	2 5/16	1.375	1 3/16	.531	1.430 1.433	.969	9/32
MS24314-32		2	2.813	1 1/16	1.995	2 5/16	1.625	1 3/16	.656	1.680 1.683	1.094	9/32

DIMENSIONS SPLINE YOKE

NOM SIZE	M ±.010	N MIN
3/4	2.875	1 7/16
1	3.125	1 3/4
1 1/4	3.312	2
1 1/2	3.750	2 1/4
1 3/4	4.438	2 5/8
2	4.625	2 3/4

(A) CANCELLED AFTER 15 AUG 1977 NO SUPERSEDING STANDARD

STATIC TORQUE MAXIMUM LB. IN.	WEIGHT JOINT TOTAL MAX LBS
1500	.20
3000	.45
7500	.80
11000	1.25
15000	2.00
20000	3.00

NAS 381 EXTERNAL INVOLUTE SPLINE DATA (c) 16/32 DIAMETRAL PITCH

NOM JOINT SIZE	NO. TEETH	PITCH DIA REF	BASE CIRCLE DIA REF	FILLET ROOT OUTSIDE DIA +.0000 -.0050	FILLET ROOT ROOT DIA +.0000 -.0010	FILLET ROOT FILLET RAD MIN	T.I.F. DIA MAX	MEASUREMENT OVER PINS MAX	MEASUREMENT OVER PINS MIN	TOOTH THICKNESS MAX EFFECTIVE	TOOTH THICKNESS MIN DIMS	MEAS. PIN DIA	ACC PITCH ERROR BETWEEN ANY TWO TEETH
3/4	11	.6875	.5954	.7500	.5625		.6250	.8395	.8574	.0977	.0946	.1200	.0022
1	13	.8125	.7036	.8750	.6875	.019	.7300	.9866	.9844	.0977	.0946	.1200	.0022
1 1/4	15	.9375	.8119	1.0000	.8125	.018	.8750	1.1131	1.1109	.0977	.0946	.1200	.0023
1 1/2	17	1.0625	.9202	1.1250	.9375	.018	1.0000	1.2392	1.2370	.0977	.0946	.1200	.0025
1 3/4	19	1.1875	1.0284	1.2500	1.0625	.017	1.1250	1.3652	1.3629	.0977	.0946	.1200	.0026
2	21	1.3125	1.1367	1.3750	1.1875	.017	1.2500	1.4909	1.4886	.0977	.0946	.1200	.0027

MAX. INVOLUTE PROFILE ERROR .0005 IN. TOOTH SURFACE PROFILE (b) MAX. LEAD ERROR .0003 IN MAX. OUT OF ROUNDNESS. .0010 IN.

(a) FEDERAL ITEMS IDENTIFICATION NO.
(b) B DIMENSION IS LENGTH OF STRAIGHT CYLINDRICAL SECTION.
(c) CENTER LINE OF PILOT HOLE (1) SHALL LIE IN A DIRECTION PARALLEL TO THE PIVOT CENTER LINE OF THE YOKE WITHIN 0°30'. CENTER LINE OF PILOT HOLE (2) SHALL LIE IN A DIRECTION PERPENDICULAR TO THE PIVOT CENTER LINE OF THE YOKE WITHIN 0°30'.
(d) CENTER LINE OF INSPECTION HOLE SHALL LIE IN A DIRECTION PARALLEL TO THE PIVOT CENTER LINE OF THE YOKE WITHIN 0°30'.
(e) PRESSURE ANGLE = 30°; ADDENDUM (BASIC) = .0313; DEDENDUM = .0395; CIRCULAR PITCH = .1963.
(f) O DIAMETER TYPICAL FOR BOTH YOKES.

MATERIAL: SEE PROCUREMENT SPECIFICATION.
FINISH: SEE PROCUREMENT SPECIFICATION.
REMOVE ALL BURRS & SHARP EDGES.

ALL DIMENSIONS TO BE MET AFTER PLATING.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64, DECIMALS ± .005.
THE TWO YOKES OF THE ASSEMBLED UNIVERSAL JOINT SHALL BE CONCENTRIC TO EACH OTHER WITHIN .013 T.I.R.
ADD -N BEFORE DASH NO. FOR JOINTS WITHOUT PREDRILLED PILOT HOLES.
EXAMPLE OF PART NOS: MS24314-12 = 3/4 IN. NOMINAL HUB WITH PREDRILLED PILOT HOLES;
MS24314-N-12 = 3/4 NOMINAL HUB WITHOUT PREDRILLED PILOT HOLES.
ALL MACHINED SURFACES MUST BE SMOOTH 125 MAX. AND FREE FROM TOOL MARKS.
"THIS DOCUMENT HAS BEEN PROMULGATE BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN, AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN."

(A) DENOTES CHANGES

APPROVED 6 FEBRUARY 1956 REVISED (A) 15 AUG 1977

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS24314
Army - OSR Navy - Bu&Ar Air Force	A - FT N - On	UNIVERSAL JOINT, ANTIFRICTION BEARING, EXTERNAL SPLINE	
PROCUREMENT SPECIFICATION MIL-U-3963	SUPERSEDES:		SHEET 1 OF 1

DD FORM 672-1
1 OCT 62

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